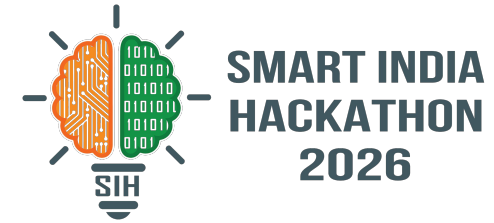
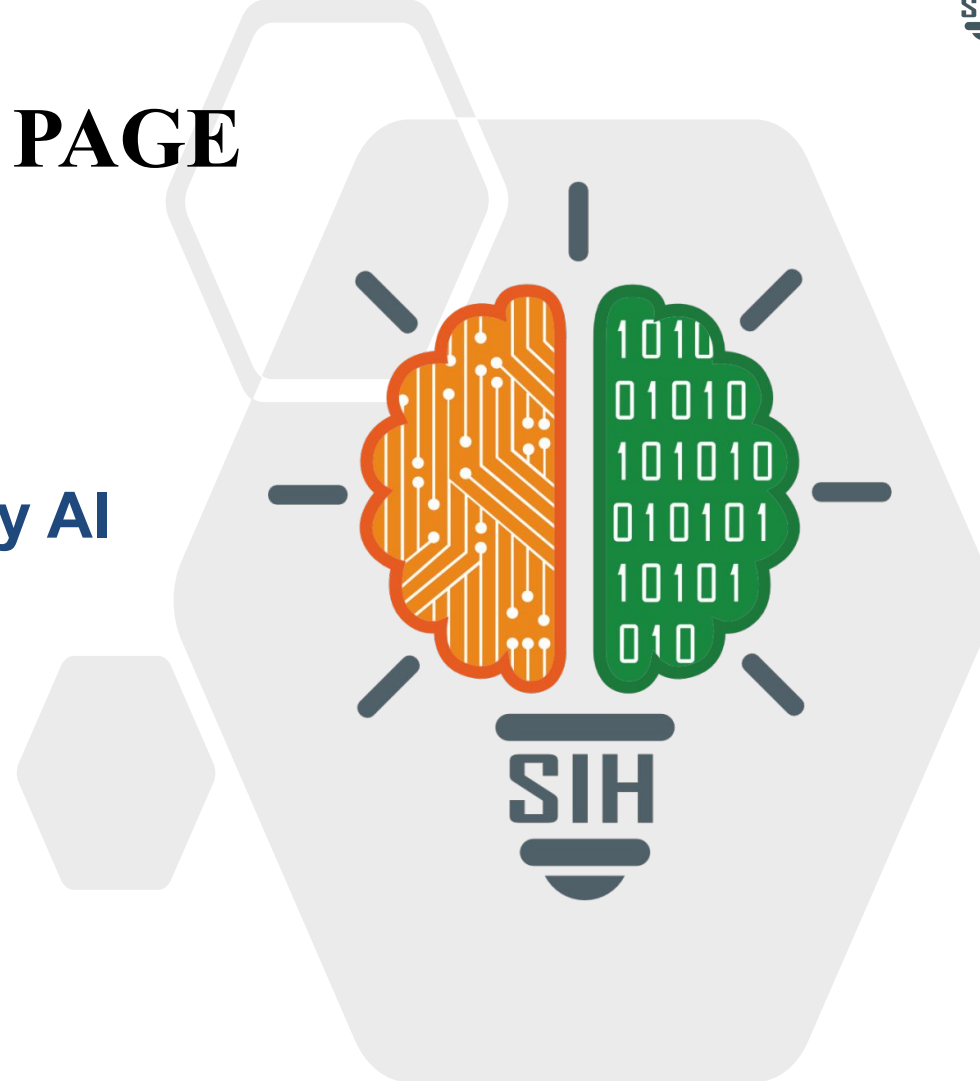


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID –SIH26167
- Problem Statement Title- SatQuery AI
- Theme- Space Technology
- PS Category- Software
- Team ID- 178961
- Team Name: Seal Team 6_SUAS26



IDEA TITLE

IDEA DESCRIPTION:

- An agentic, query-driven VLM assistant that turns natural-language questions into evidence-grounded answers from satellite imagery.
- Accepts single images, co-registered optical–SAR pairs, and bi-temporal pairs (GeoTIFF/TIFF).
- An agentic controller interprets the query, validates inputs, and routes work to specialist remote-sensing models.
- Delivers VQA, captioning, text-guided grounding, change understanding, and optical–SAR fusion from one interface.
- Every response returns visual evidence, a confidence score, and an auditable execution summary.

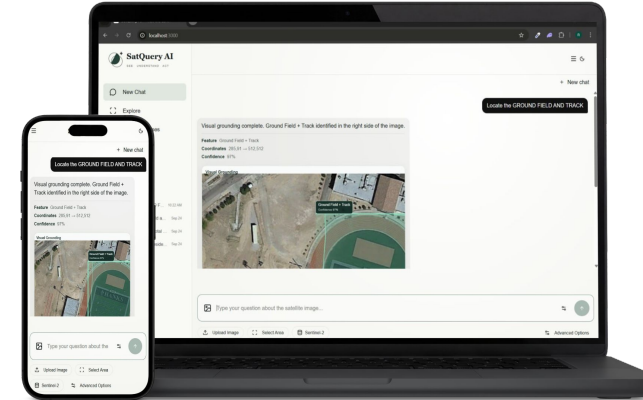


[Video Demonstration](#)



[Project Link](#)

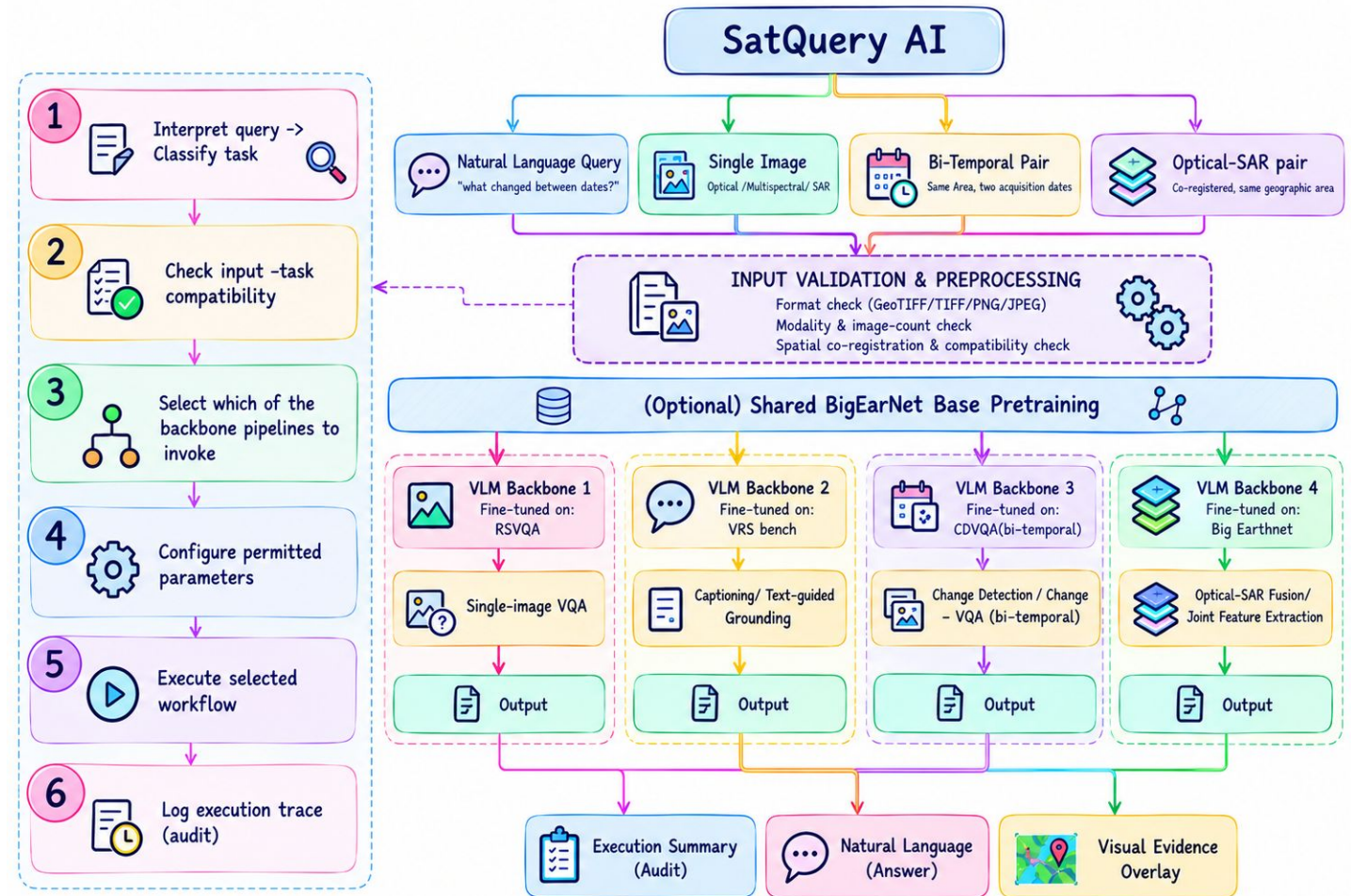
Web & Mobile Application Interface



NOVELTY:

- Agentic by design: dynamically selects and sequences specialist models per query, unlike single-task RS tools.
- True cross-modal + multitemporal reasoning — jointly interprets optical, SAR, and time-series pairs, not just one image.
- Domain-adapted, not generic: fine-tuned on BigEarthNet and RS-specific benchmarks instead of relying on an off-the-shelf VLM.
- Transparent and auditable — every answer ships with an execution trace of tasks, models, and parameters used.
- Built for non-experts: natural-language access removes the need for GIS or model-selection expertise.

Frontend	Next.js
Backend	Python, Fastapi, Uvicorn
AI/deep learning	Pytorch, Huggingface, unsloth
Model Ecosystem	Qwen2.5-VL , LLama, SARMAE
Database	PostgreSQL, VectorDB
GPU Acceleration	NVIDIA CUDA
Version Control	Git & Github



Feasibility

Technical Feasibility

- Pretrained VLMs (QWEN 2.5) are readily adaptable via fine-tuning.
- BigEarthNet, VRSBench, RSVQA, CDVQA provide ready domain data.
- Modular agent architecture supports incremental development.

Market & Usability

- Rising demand for accessible RS analytics across agri, disaster and urban domains.
- NL interface removes the GIS/model-selection expertise barrier.
- Directly aligned with ISRO/SAC evaluation and operational needs.

High-Feasibility Indicators

- Open-source benchmark datasets reduce data-collection risk.
- Proven baseline architectures exist for each specialist task.
- Loosely-coupled modules lower integration and testing risk.

Challenges

Challenge: Scarce paired optical–SAR & bi-temporal data

Solution : Leverage BigEarthNet and cdvqa with augmentation & self-supervised pretraining

Challenge: Domain gap between generic VLM and RS imagery

Solution : Fine-tune on bigearthnet,VRSBench, RSVQA and CDVQA instruction data

Challenge: SAR speckle noise & modality heterogeneity

Solution: Dedicated SAR encoder with denoising preprocessing

Challenge : Reliable task routing for ambiguous queries

Solution : Hybrid rule + LLM controller with confidence thresholds & fallback

Challenge: Large GeoTIFF sizes & compute limits

Solution: Patch/tile-based inference with GPU-optimised caching

Challenge: Evaluation on undisclosed ISRO/SAC dataset

Solution: Generalisable design validated on public benchmarks first

Potential Impact

Farmers & Agri-Agencies



- Fast crop and land-cover monitoring
- NoGIS expertise required

Disaster-Response Team



- Rapid flood, fire and damage assesment
- Through natural-language queries

Urban Infra planners



- Bi-temporal imagery for tracking built-up growth
- Infrastructure and land-use change analysis

Research Institutions



- Unified platform for evaluating deploying and extending RS-VLMs
- Supports research and operational deployment

Benefits

OPERATIONAL



- Single unified platform replacing multiple isolated RS tools
- Faster decisions via natural-language queries and automatic orchestration
- Adjustable execution trace builds user trust and adoption

Scientific



- Advances multimodal (optical + SAR) and mulitemporal RS-VLM research
- Benchmarked on BigEartNet, VRSBench, RSVQ and CDVQA
- Reusable specialist-model registry extendable to future RS tasks

Economic



- Reduces dependence on specialist GIS/RS workflows
- Scalable GPU/cloud deployment lowers cost per query
- Enables faster, data-driven policy and resource-allocation decisions

[BigEarthNet.txt: A Large-Scale Multi-Sensor Image-Text Dataset and Benchmark for Earth Observation](#)

[RSVQA: Visual Question Answering for Remote Sensing Data](#)

[VRSBench: A Versatile Vision-Language Benchmark Dataset for Remote Sensing Image Understanding](#)

[ThinkGeo: Evaluating Tool-Augmented Agents for Remote Sensing Tasks](#)

[Deep Learning in Multimodal Remote Sensing Data Fusion: A Comprehensive Review](#)

[Change Detection Meets Visual Question Answering \(CDVQA\)](#)

[RSVG: Exploring Data and Models for Visual Grounding on Remote Sensing Data](#)

[Multi-Agent Geospatial Copilots for Remote Sensing Workflows \(GeoLLM-Squad\)](#)

[SAR Strikes Back: A New Hope for RSVQA](#)